

Drilling Fundamentals II

Introduction to Drilling Technology

Volker Wittig

International Geothermal Centre Bochum, Germany

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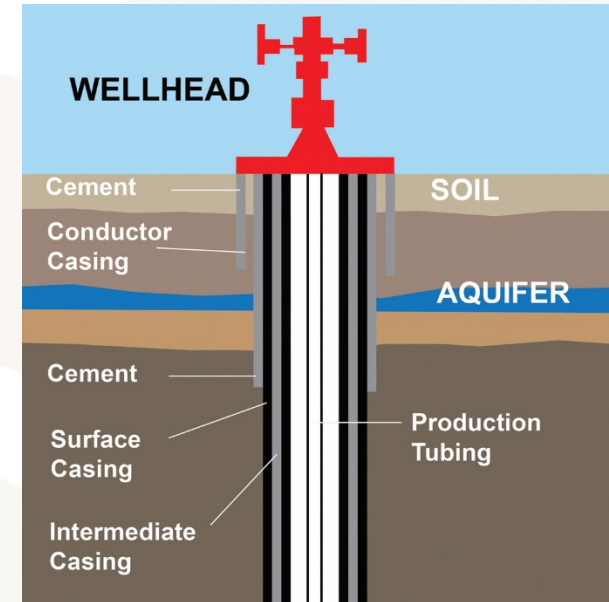
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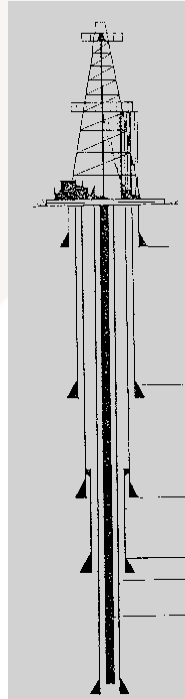
Casing

Main objectives of casing program

- Prevent collapsing, bursting, buckling and other casing damages,
- Support drilling and anchor wellheads,
- Safely contain well fluids,
- Control contamination of subsurface aquifers,
- Counter circulation losses during drilling,
- Protect integrity of the well against corrosion, erosion and fracturing.



Casing program



Conductor / surface Pipe (18" - 30")

anchor / structural Casing (13" - 20")

Intermediate Casing

Production Casing (5" - 7")

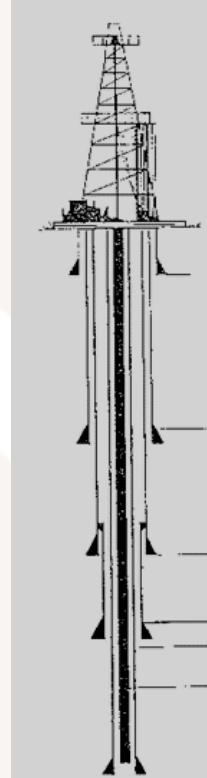
Liner (5" - 7")

Production Tubing (< 5")

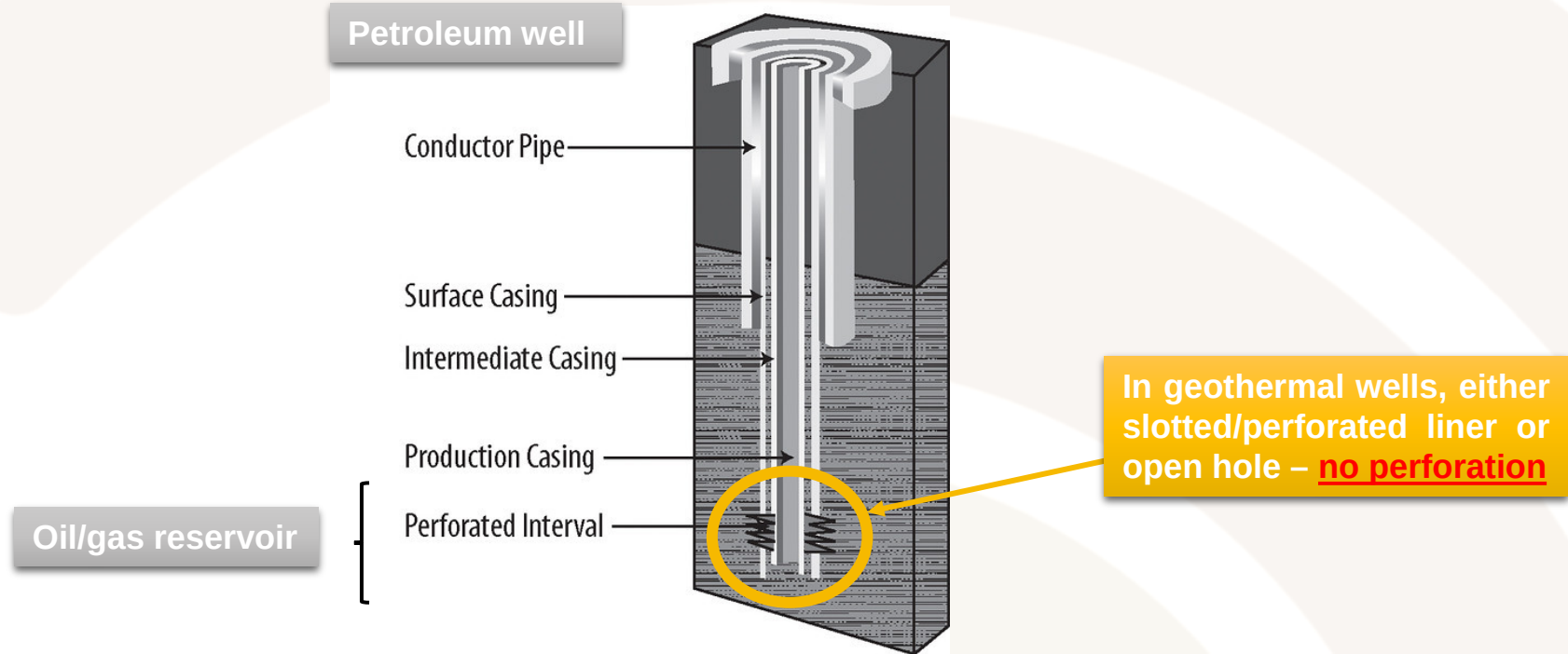


Casing program

Casing string	Main roles
Conductor / surface pipe	To prevent the hole and drill pad from caving into the wellbore
Anchor / structural casing	Support of the BOP and the final production wellhead; should be deep enough to take up all eventual forces / pressures / blow-out; also protects near surface freshwater aquifers and provides minimal pressure integrity
Intermediate casing	Provides protection against caving of weak or abnormally pressured formations and enables the use of drilling fluids of different density necessary for the control of lower formations as well as anchors the wellhead
Production casing	Isolate the zone containing natural gas from other subsurface formations; pump hydraulic fracturing fluids into the producing formation without contacting other formations along the wellbore.
Liner	Prevents loose formation from collapse; does not reach the surface – cost reduction, may serve as corrosion/scaling indication tool
Production Tubing	Allows for production of the reservoir fluids and installation of pump if necessary, protects wellbore casing from wear, tear, corrosion, and deposition of by-products, such as sand / silt, paraffins, and asphaltenes



Typical hydrocarbon casing program



Hole and casing sizes

Hole Size

36"

Structural casing

26"

Conductor string

17 1/2

Surface pipe

12 1/4

Intermediate string

8 3/4

Production Liner

Casing Size

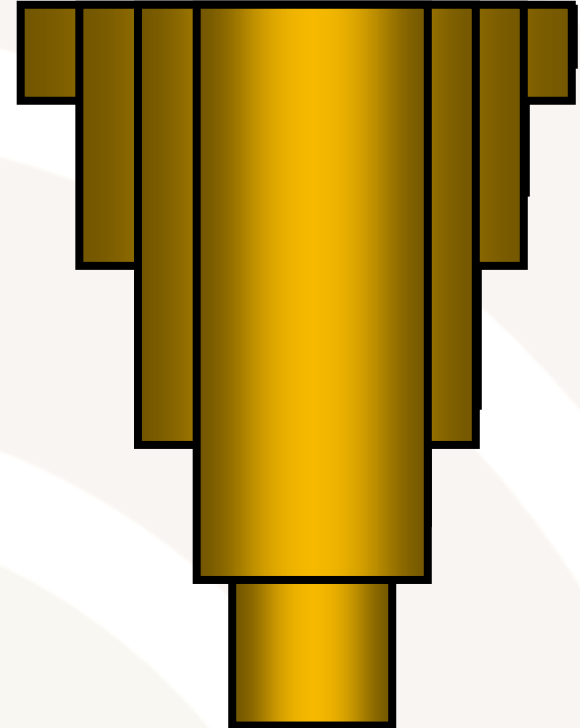
30"

20"

13 3/8

9 5/8

7



Casing design + setting depths

Before drilling

- Geology + Lithological profiles
- Seismic data (2D & 3D)
- Offset well data



Fracture Gradient Prediction
Pore Pressure Prediction

During drilling:

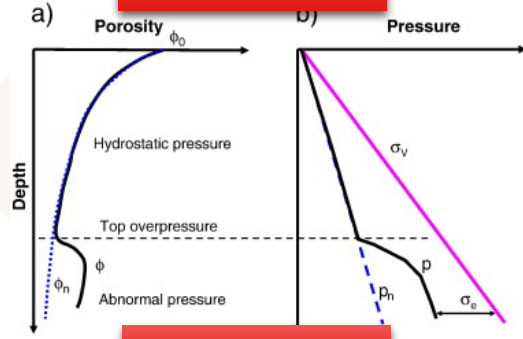
- Measurement while Drilling (MWD → data of the drilling process)
- ECD → equivalent circulation density → estimating formation pore and “Frac” pressures
- Kicks, porosity
- Cuttings
- Mud logging analysis

After drilling:

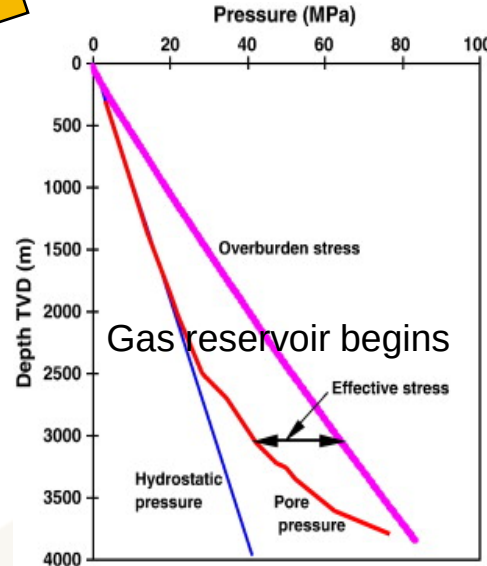
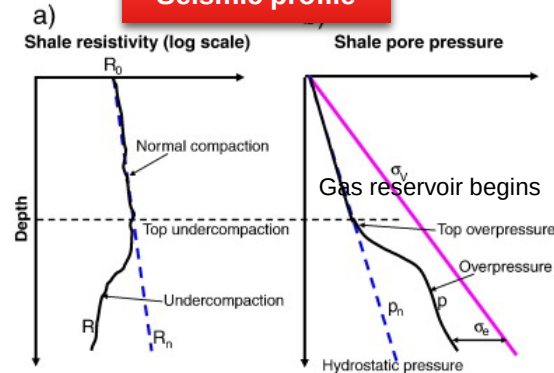
- Well logging (Resistivity, Conductivity, Sonic, Density)
- Bottomhole pressure surveys

Casing setting depth

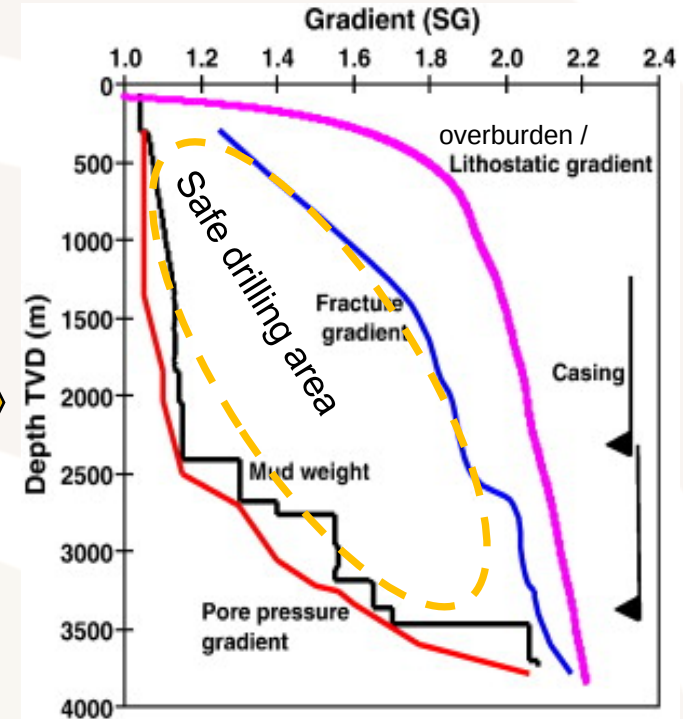
Pore pressure



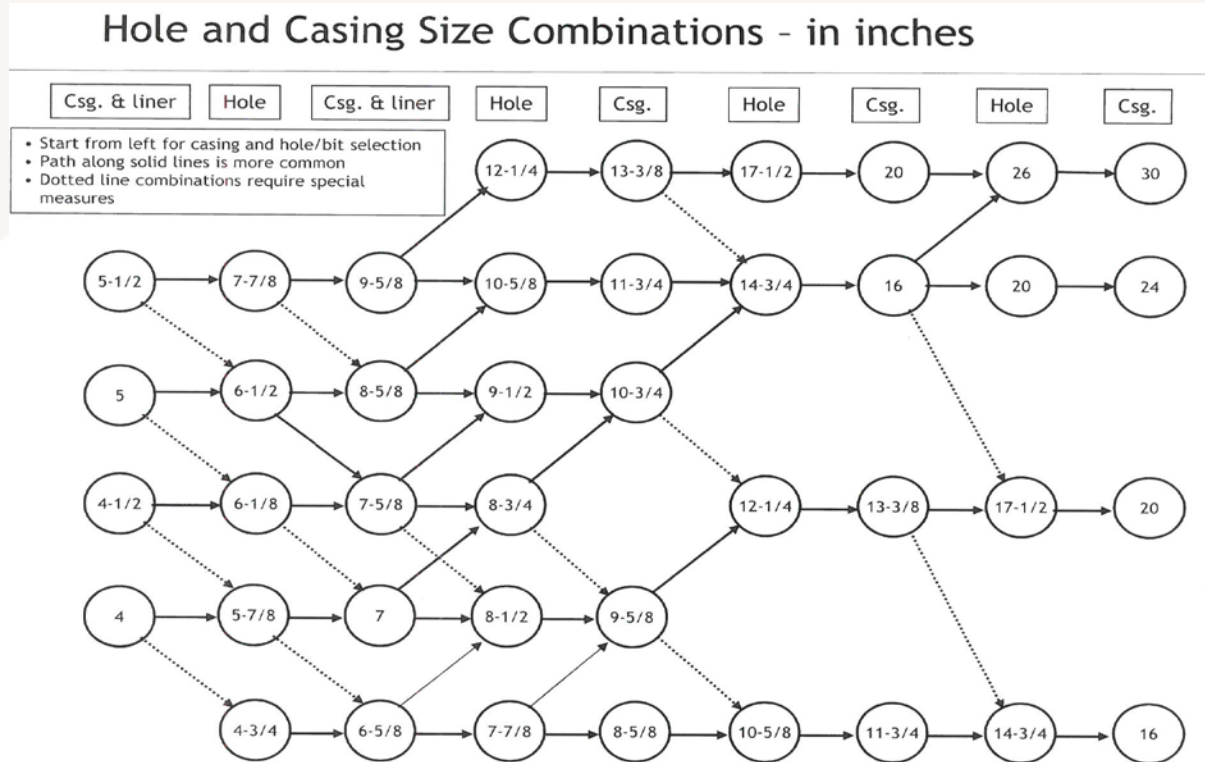
Seismic profile



$$\sigma_{\text{eff}} = \sigma_{\text{ov}} - \alpha * p_{\text{pore}}$$



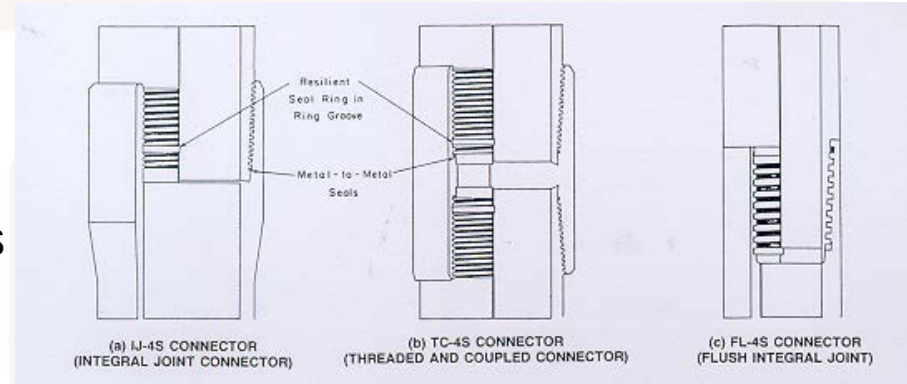
Hole and casing sizes (API)



Casing threads and couplings

Connections, examples:

- Integral joint connector
- Threaded and coupled connectors



Different API thread types:

- API - round thread - coupled connection
- API buttress thread - coupled connection
- API - Extreme - Line - integral connection

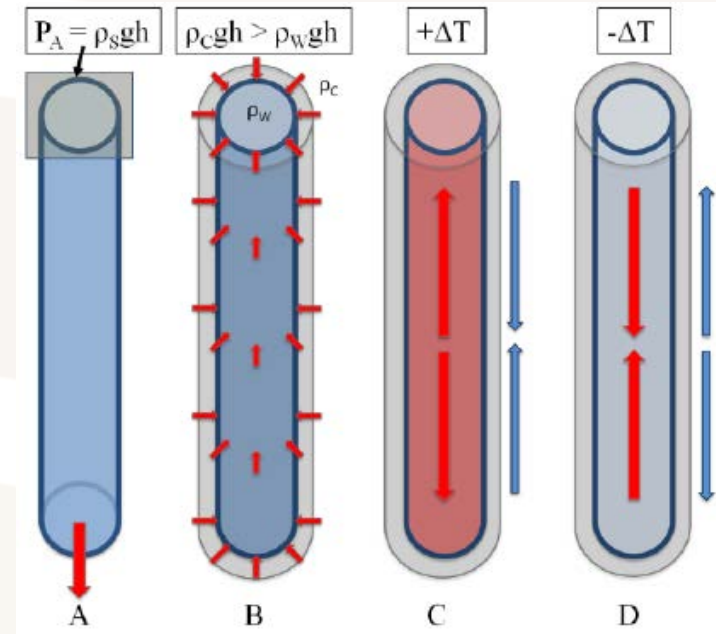
Casing loads

The casing loads that are typically used in preliminary casing design are:

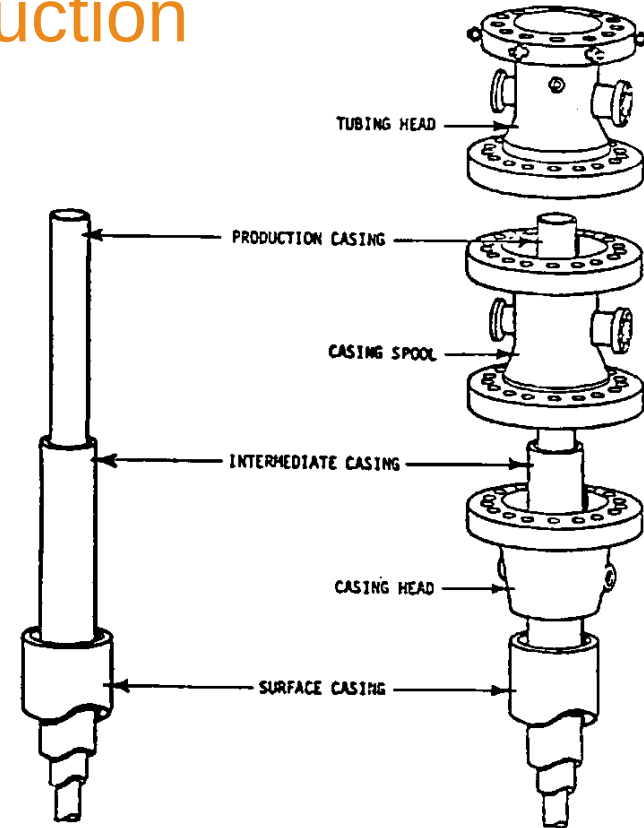
- **External Pressure Loads (Collapse pressure)**
- **Internal Pressure Loads (Burst pressure)**
- **Mechanical Loads**
- **Thermal Loads and Temperature Effects**

Mechanical loads are associated with:

- Casing hanging weight
- Shock loads during running
- Packer loads during production and workovers
- Liner hanger loads



Wellhead for control and production



Additional casing equipment

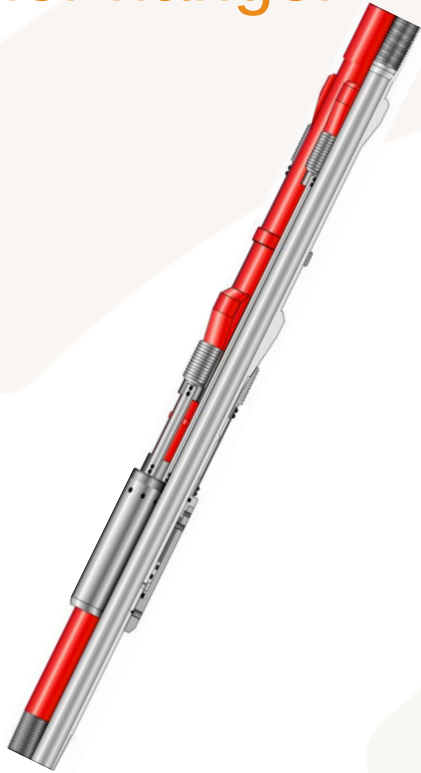
- Centralizers
- Casing shoe
- Float collar
- Plugs
- Liner hangers

During workover:

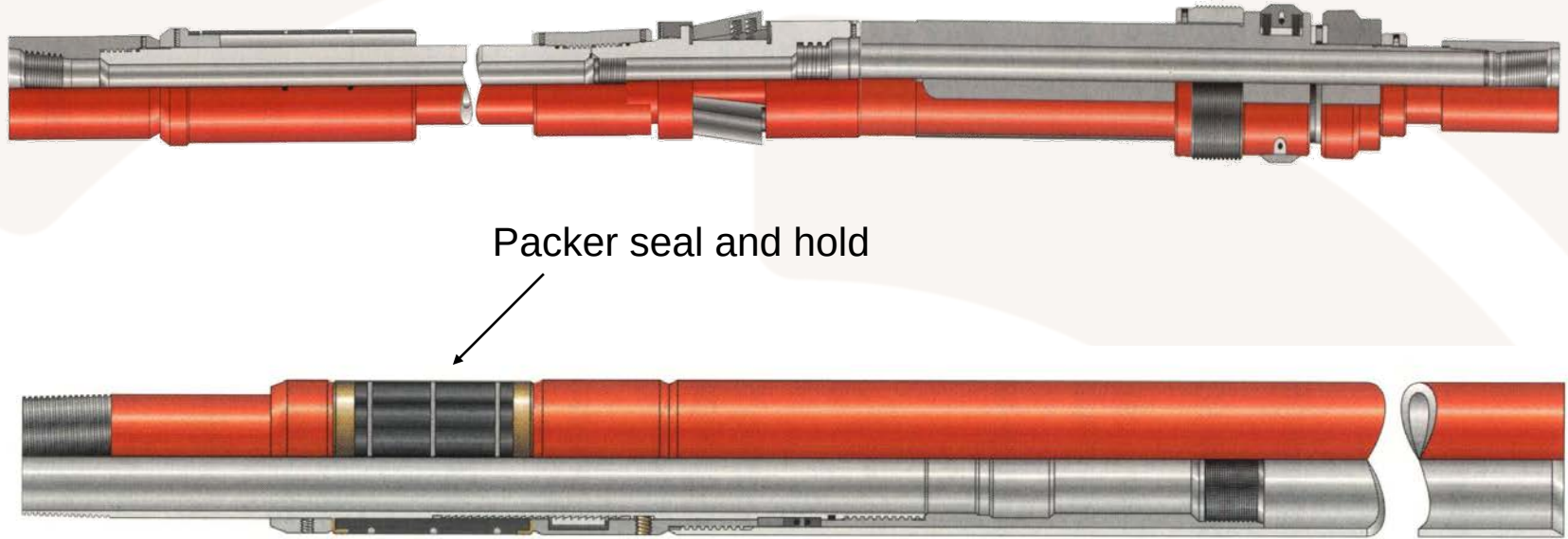
- Scratchers
- Chemicals



Liner hanger



Liner hanger system



Casing running

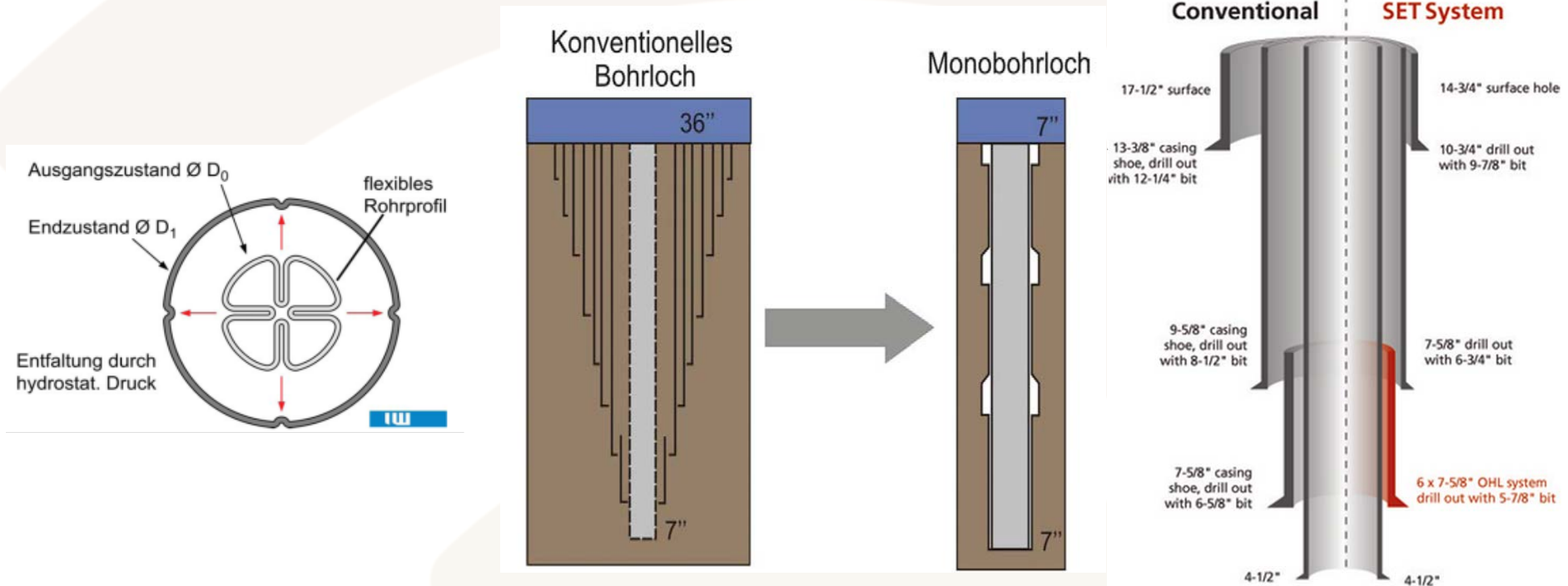
Time consuming
Tough job



Casing running : makeup torque

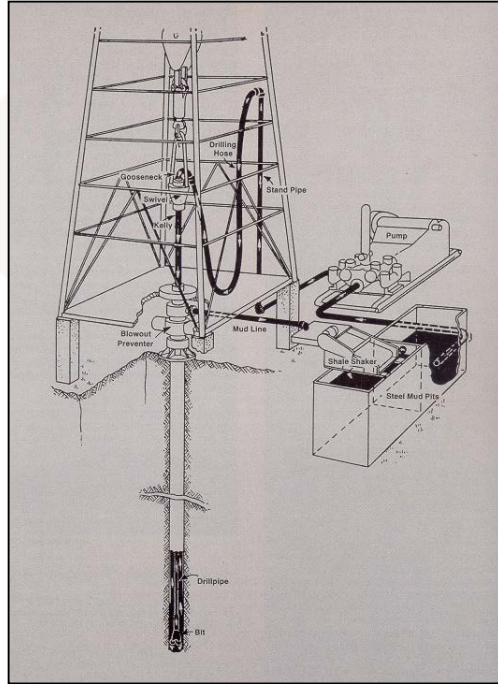


New technology – expandable liner / casing

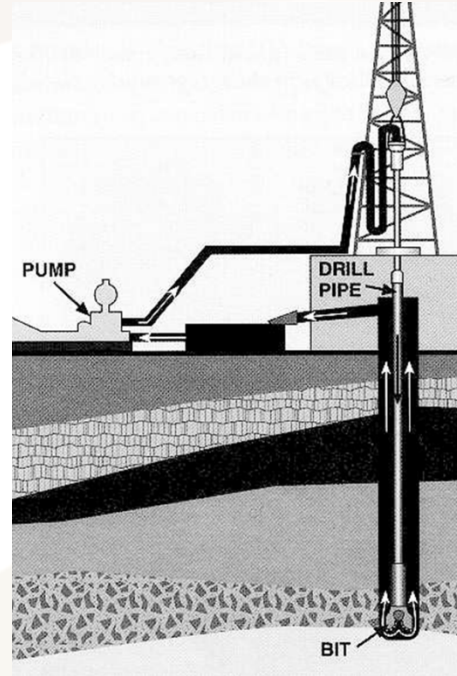


Drilling Fluid / mud system

Mud circulation system



Typical mud circulating system on a rotary drill rig (Moore, W.W., 1981: p.138)



Drilling fluid / mud

Drilling fluids / mud are circulating liquids and gases inside the wellbore during drilling operations.

Drilling fluid may be based on water or oil (mainly petroleum wells)

Drilling fluid is different from:

- Working and killing fluids
- Annulus fluids
- Cementation fluids



Functions of drilling fluid / mud

- 1. Transport cutting and dispose to surface** – The drilling fluid brings the drilled material to the ground surface either by mud rheology and velocity.
- 2. Clean drill bits** – As drilling fluid exits the bit jets, fluid velocity removes cutting from the bit teeth and bit body. This prevents bit ball up situation.
- 3. Provide hydrostatic pressure to control well while drilling** – Hydrostatic pressure provided from drilling fluid is the primary well control. Mud weight should be high enough to control formation pressure while drilling.
- 4. Prevent excessive mud loss** – While drilling, clay particle will form a thin layer over porous zones called “**mud cake**”. Mud cake acts as barrier to prevent excessive drilling fluid loss into formation and provides wellbore stability.
- 5. Prevent formation damage by using reservoir drill-in fluid** – While drilling long reach zone in horizontal wells, the special drilling fluid will be utilized in order to prevent formation damage.
- 6. Provide hydraulic energy to downhole BHA as mud motor, measuring while drilling (MWD), logging while drilling (LWD), etc** – Without enough hydraulic power, downhole tool will not be properly operated, hence, drilling fluid plays essential role to provide power to sophisticated downhole tool.
- 7. Facilitate downhole measurement as open hole logging, MWD, LWD, mud logging, etc** – Mud will assist tool to measure everything downhole.
- 8. Lubricate drill string and BHA and cool the bit. The drill bit and BHA become hot due to friction during the drilling process.** When the drilling fluid passes through the bit and exits the jets/nozzles, some extra heat is removed via mud.

Some drilling fluid components

Main base: fresh water, salt water

Main additive: bentonite clay

Oil based muds, OBM : crude or Diesel oil

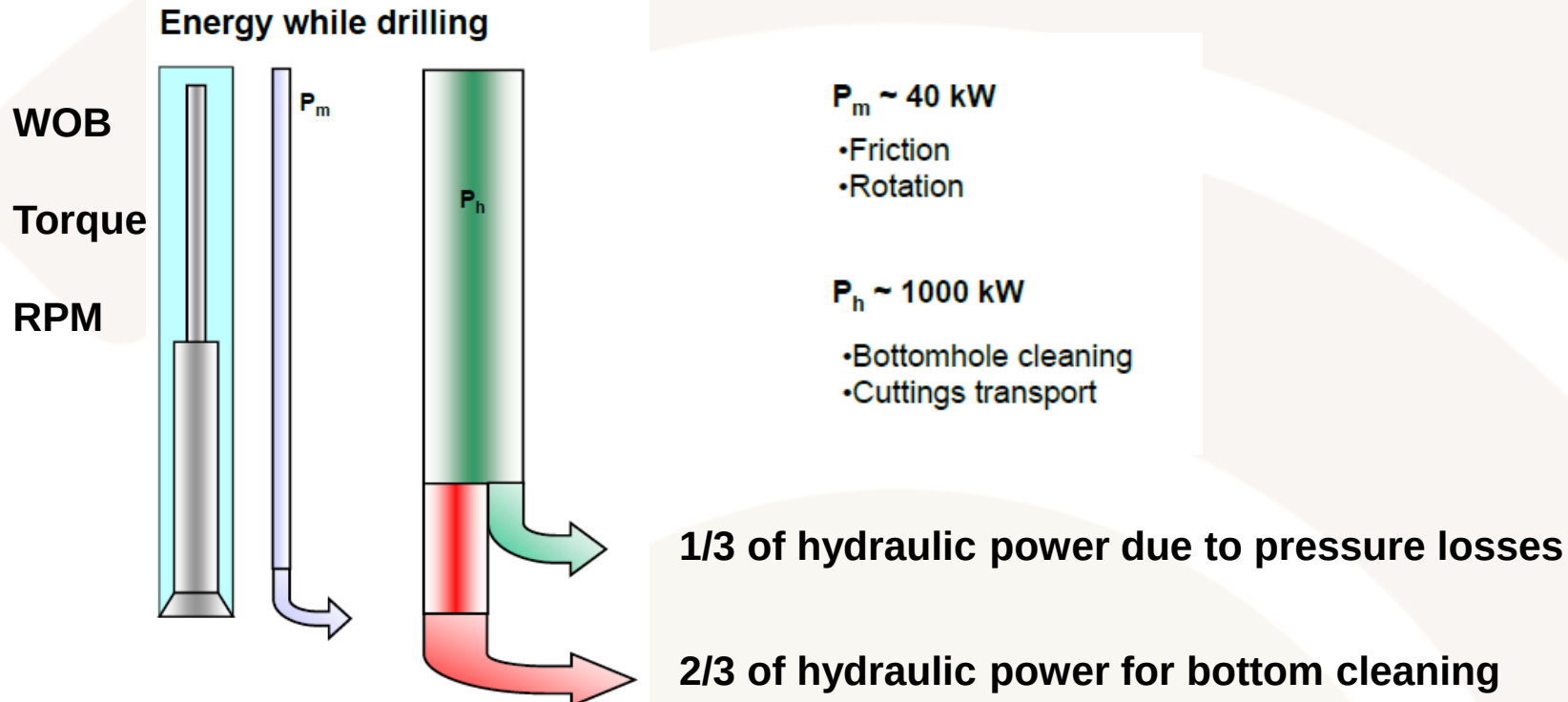
Water loss control, filter cake thickness reduction: starch (up to 80 ° C), carboxymethylcellulose (CMC, up to 150 ° C); synthetic polymers

Viscosity reducing agents: phosphates, tannates, humates, lignosulfates

Weighing agents: barite, salt, calcium carbonate, chalk

Other additives for high temp. Etc.: Electrolytes, additives to reduce flushing losses, inhibitors, defoamers, pH regulators

Energy for drilling : > 90 % hydraulic energy



Differential pressures – borehole control

Drill mud density is acting as a force, which pushes against the formation pressure



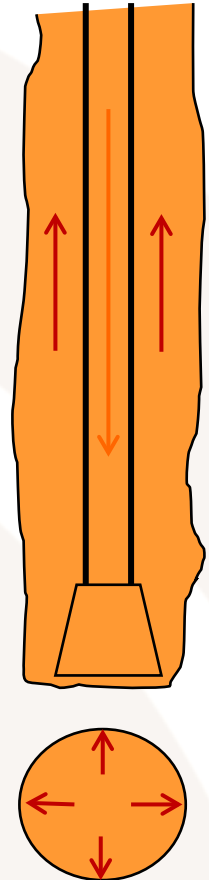
$$P = \rho_{\text{mud}} * g * h$$



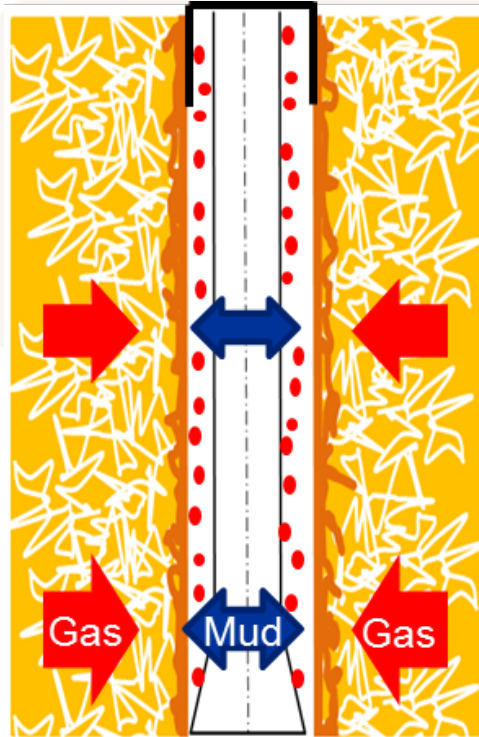
$$P = \rho_{\text{cutting loaded mud}} * g * h$$

ECD= Equivalent Circulating Density > $\rho_{\text{cutting loaded mud}}$
(includes ΔP = pressure loss in annulus while pumping)

Drill mud density is acting as radial pressure to stabilize bore hole



Differential pressures – balancing mud and formation pressure



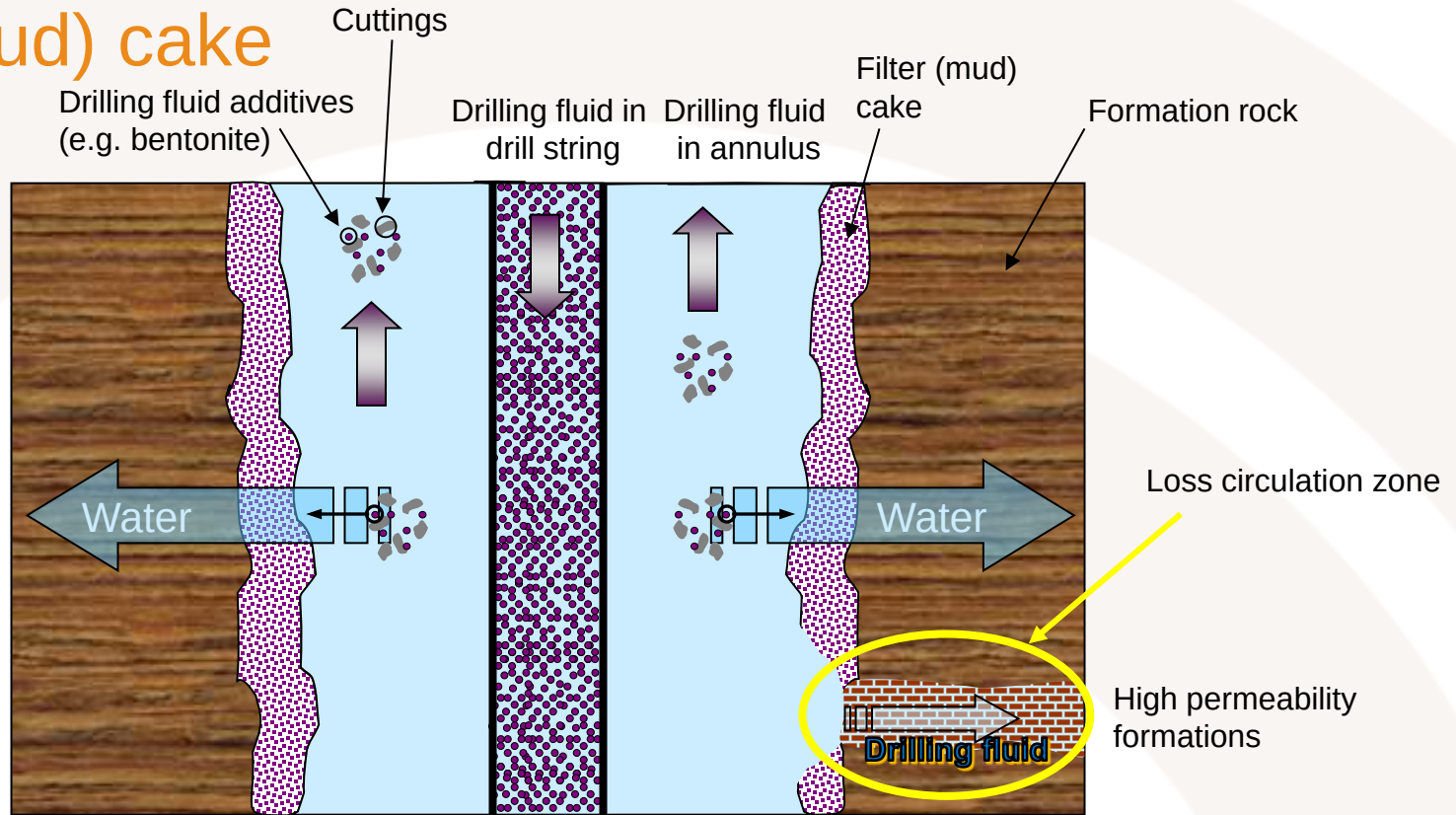
Balanced Drilling:
Mud Pressure = Formation pressure

Underbalanced Drilling:
Mud Pressure < Formation pressure

Overbalanced Drilling:
Mud Pressure > Formation pressure

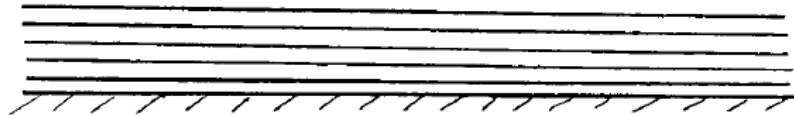
Overbalanced - Safest and most common way

Filter (mud) cake



Basic principles of drilling fluid flow – laminar flow

- Flow pattern is linear (no radial flow)
- Velocity at wall is ZERO
- Produces minimal hole erosion
- Mud properties strongly affect pressure losses
- Preferred flow type for annulus (in vertical wells)
- Laminar flow is sometimes referred to as sheet flow, or layered flow:



Basic principles of drilling fluid flow – turbulent flow

Turbulent drilling fluid flow:

- Fluid molecules (and particles) with high and low particle velocity
- Collision and change of impulses
- Velocity distance = medium particle velocity in fluid flow direction
- Distance of velocity highly consistent, therefore smooth velocity profile
- Close to wall of pipes vertical motion of fluid and particles --> changeover laminar flow
- Thickness of the „laminar bed“ depending on the degree of the turbulence

At critical Reynolds number
laminar changes to turbulent flow
i.e. 2000 to 3000 (2,300)

Drilling methods

Turbulent fluid flow **not** wanted:

- Pressure loss
- Drill pipes abrasion
- Wellbore erosion

Turbulent fluid flow **wanted**:

- mud cake erosion

Mud flow rate and uphole velocity

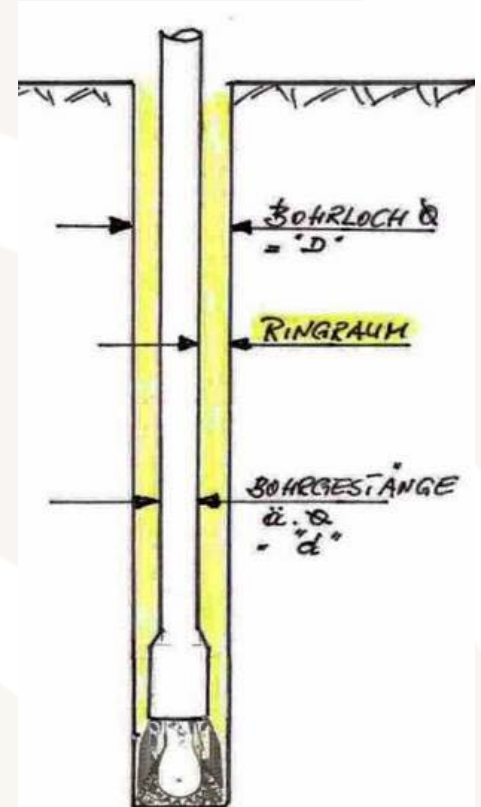
uphole velocity

$$V_{Ring} = \frac{Q}{A_{Ring}} = \frac{m^3}{min \times m^2} = \frac{m}{min}$$

Area of the annular space

$$A = [D^2 - d^2] \times 0,785$$

> 0,4 m / sec in outer annulus



Drilling fluid losses

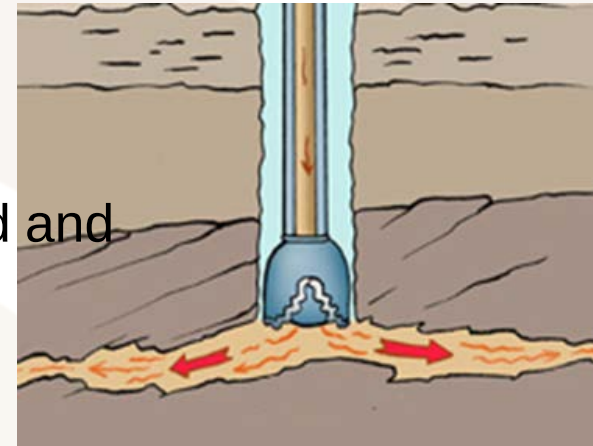
Severity of drilling fluid losses:

- Small losses from 0 to about
- Partial losses up to appr.
- Total losses up to the loss zone

1-2 m³ / hr (0.3 – 0.6 l/s)
50 m³ / hr (14 l/s)

Reasons for drilling fluid losses:

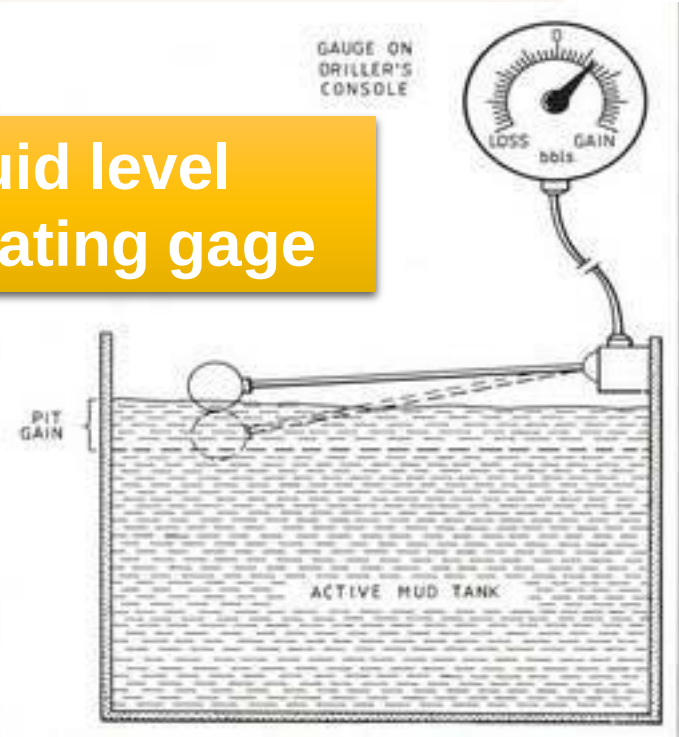
- unconsolidated, highly permeable formations (sand and gravel layers)
- natural fractured zones in the rock formations
- Fractured or fissured formations



Volume control of drilling mud

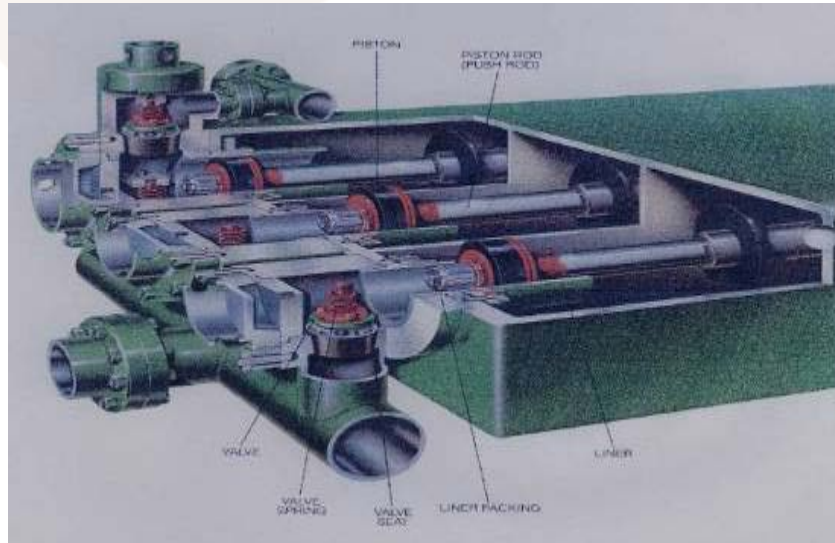
Detection of drilling fluid flow into the borehole and drilling fluid circulation losses into the surrounding formations.

Fluid level floating gage

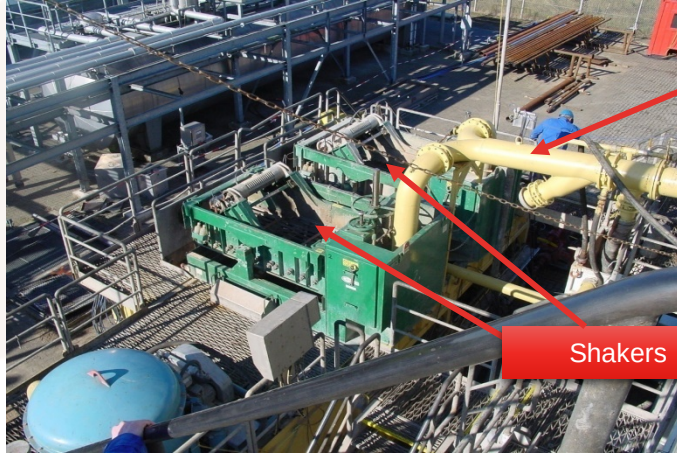


Mud pumps

triplex piston pumps with 3 pistons parallel (120° offset) with seals / sealing package in the piston are an industry standard



Mud circulating system



Shakers

Mud and cuttings
come out of the
flow line



Desander



Desilter



Centrifuge



Mud circulating system



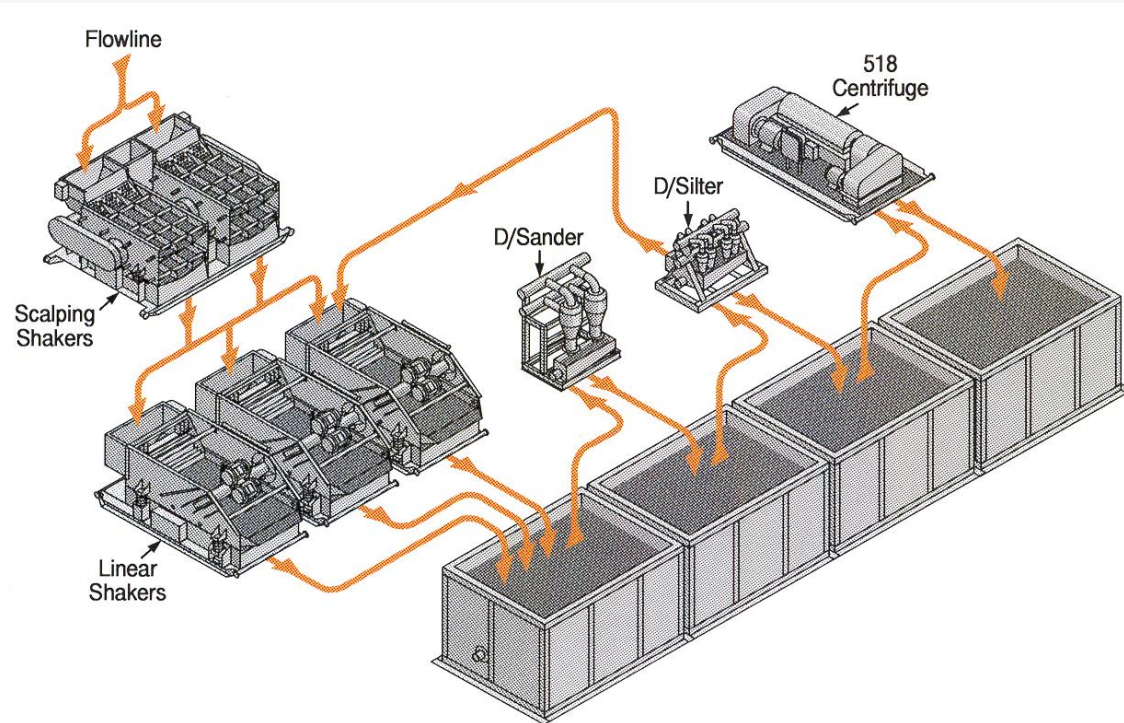
Other mud pumps - Screw type pumps / PDM

Screw pump (or Positive Displacement Pump):

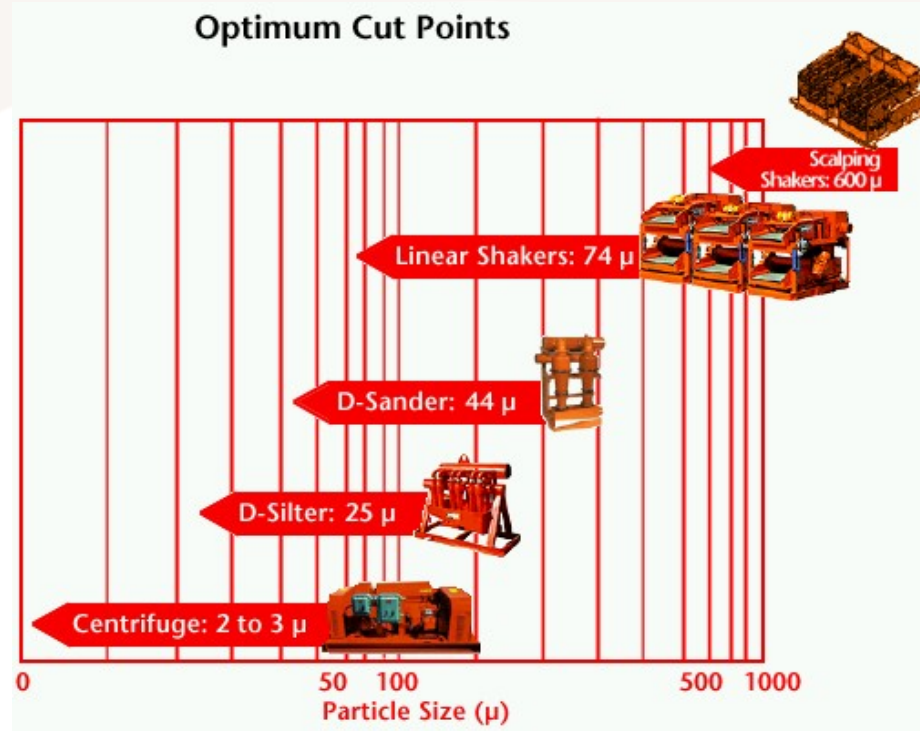
- high flow, low pressure (< 25 bars – 350 PSI)
- inexpensive



Mud / drilling fluid conditioning system



Drilling fluid preparation



Shakers

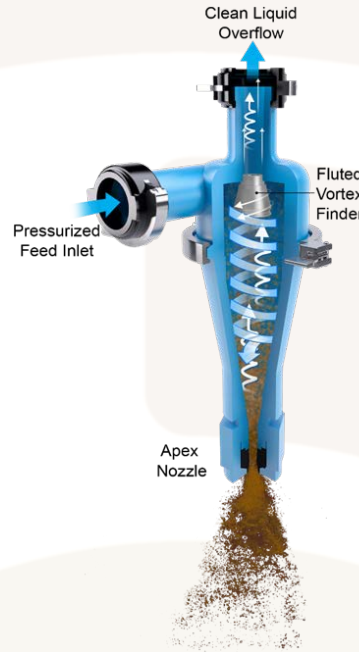
Gravity separation using silt, sand and shale shakers



Drilling fluid recycling



Desander

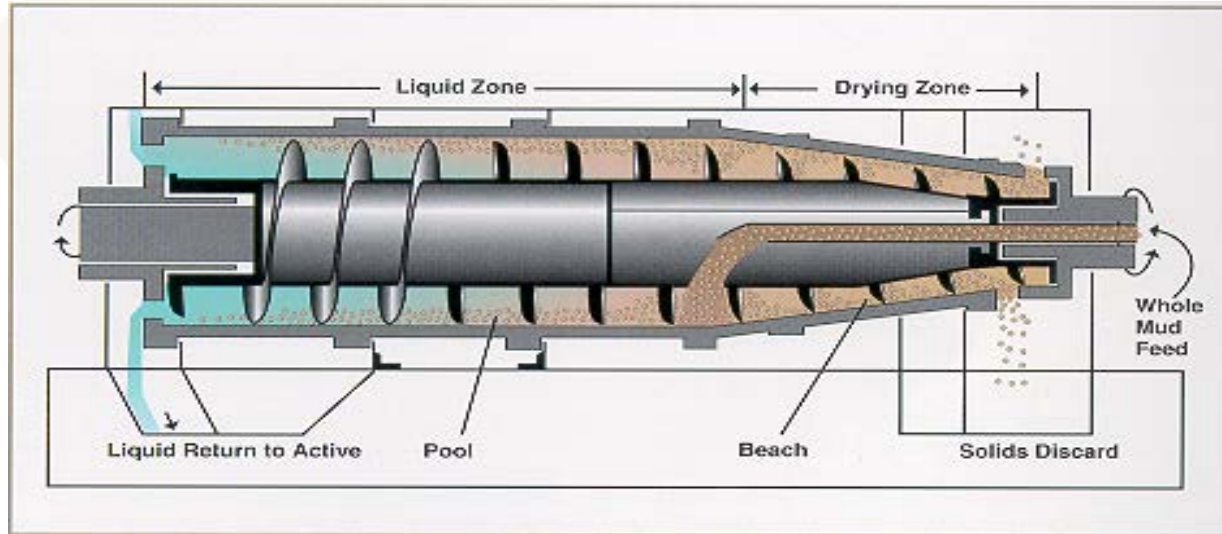


Hydrocyclon



Desilter

Drilling fluid preparation - centrifuge



Discharge from centrifuges



Drilling fluid properties

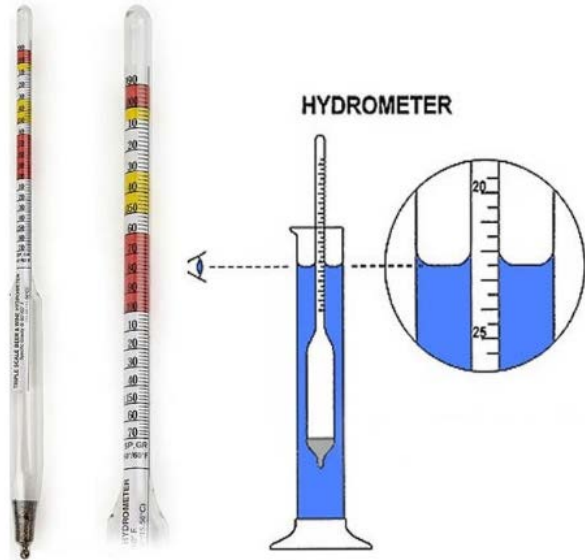
Commonly measured parameters:

- spec. weight (density)
- viscosity
- gel intensity
- filtration
- pH
- clay content
- salt content
- Grade
- Mud cake / filter press

On-site drilling mud laboratory



Drilling fluid density measurement tools

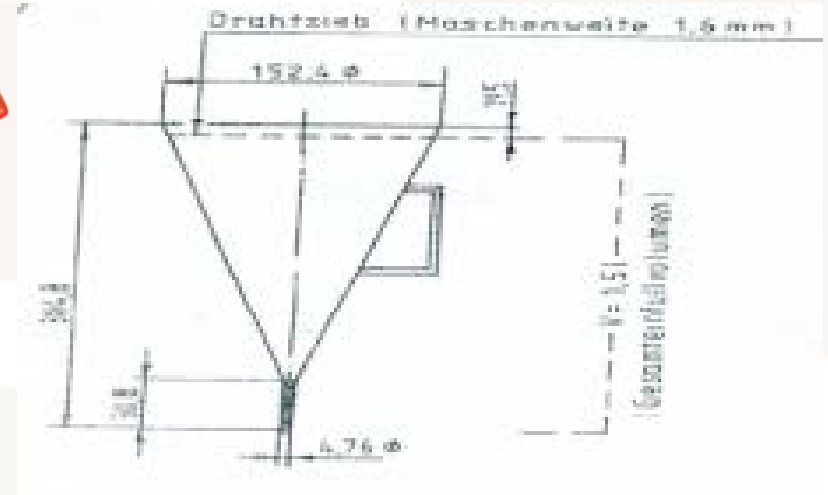


Hydrometer



Baroid drilling mud weight (scale)

Drilling fluid viscosity measurements – Marsh Funnel



Drilling fluid control equipment

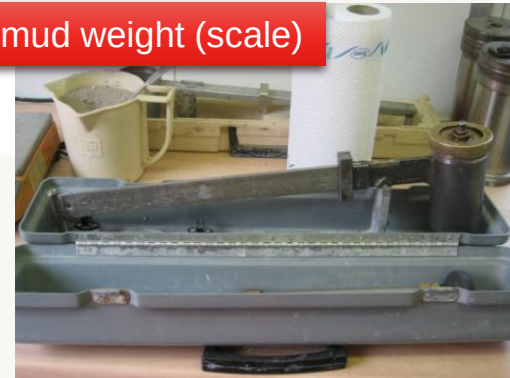
Fann viscometer



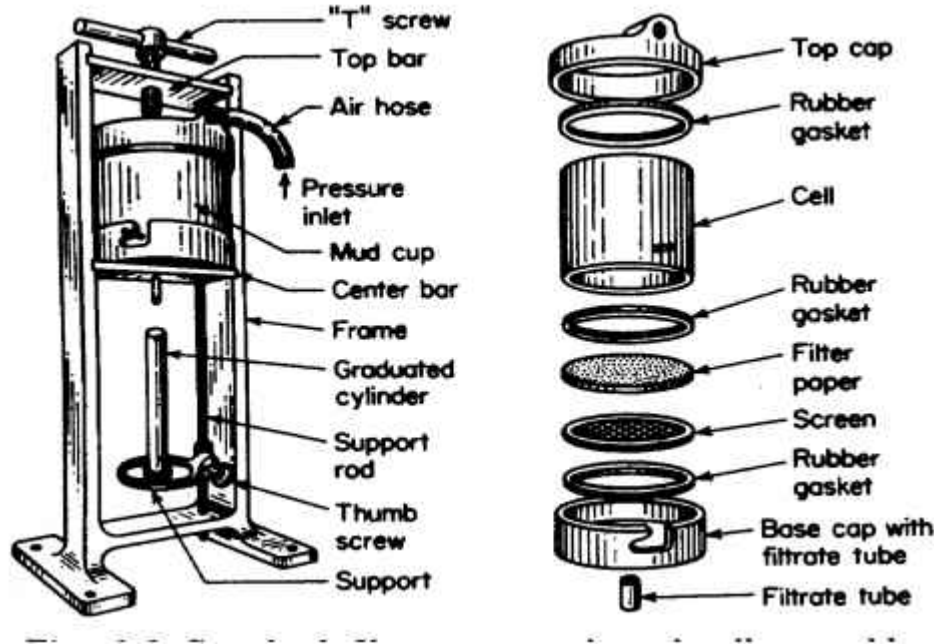
API press



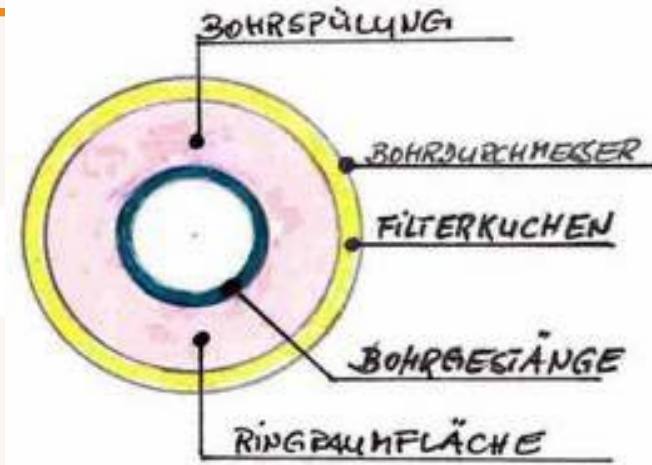
mud weight (scale)



Filtration and mud (filter) cake



Filtration is measured in ml of drilling fluid per second



Mud / filter cake

pH

pH of 7.0 is neutral

pH lower than 7.0 is acidic

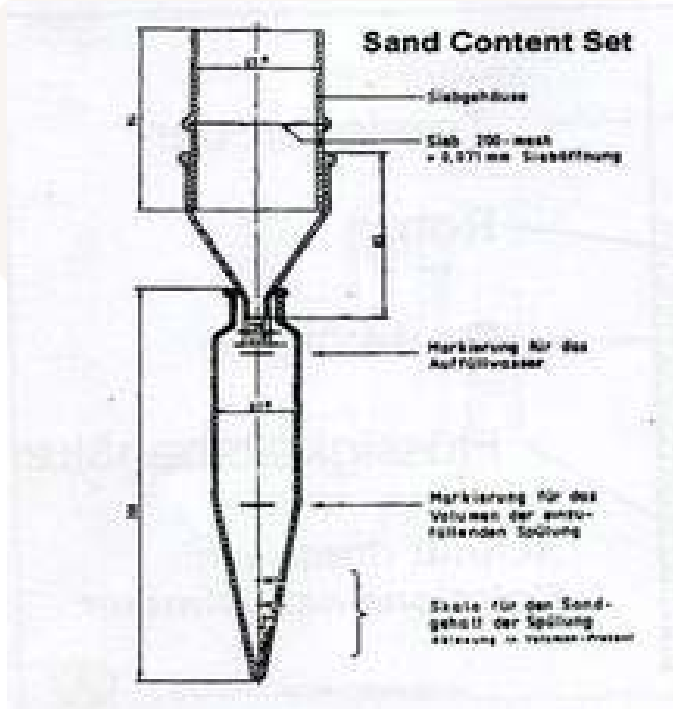
pH higher than 7.0 is more alkaline

Drilling fluid additives were developed to be mixed with water with a pH level from 8.5 to 10 in order for the required chemical reaction to occur and to provide proper function.



pH-meter

Sand content set



Measurements in % or ml

Drilling fluid measurement units + values

Spülungsdaten	Symbol	Einheit	Meßgerät oder -Verfahren
1. Dichte	ρ	kg/l	Hydrometer oder Spülungswaage
2. Auslaufzeit	az	s	Marsh-Trichter
3. Restauslaufzeit	raz	s	Marsh-Trichter
4. Austrittstemperatur	t	°C	Thermometer
5. Wasserabgabezeit	waz	s	Ringapparat
6. Wasserverlust	wv	ml	Filterpresse
7. Filterkuchendicke	fk	mm	Dickenmeßg
7a. Filterkuchentextur	fkt	(keine)	Qualitativ mi
8. pH-Wert Filtrat	ph	(keine)	Indikatorstäb
8a. pH-Wert Spülung	ph	(keine)	Indikatorstäb

Dichte	ρ	abhängig von den g Formationen
Auslaufzeit	az	von 38 s – 45 / 50 s
Restauslaufzeit	rz	von 30 s – 70 / 80 s
raz / az		≥ 0,80 – 0,90
Wasserabgabezeit	s	400 s – 1.800 s
Wasserverlust	wv	20 ml - 5 ml
pH – Wert		mind. 7 -9
Sandgehalt	sd	≤ 2 %

Cementation

Cementation

Main tasks of primary cementations:

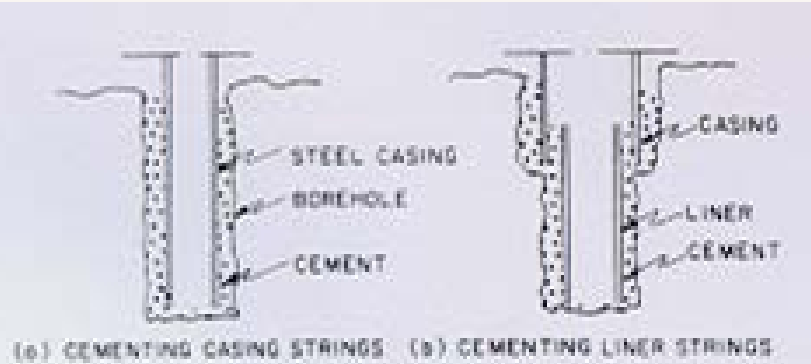
- Fluid losses
- Formation collapse
- Set and cement casing strings
- Well abandonment
- Preventing contamination of freshwater horizons
- Corrosion protection for the outer wall of the casing

Factors that are essential for good annulus cementation:

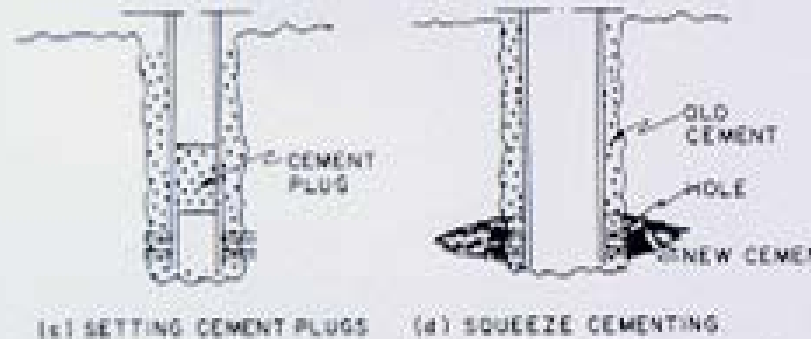
- Dimensional accuracy of the borehole
- Compatibility (tolerance) between drilling fluid and cement
- Centering of the casing
- Sufficient spacer volume / separation between drilling fluid and cement
- Running in tubing before cementation and during cementation to the annulus in order to fill the annulus as best as possible

Cementation types

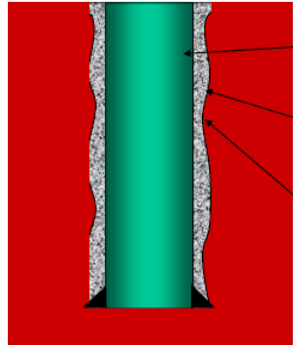
Casing cementation



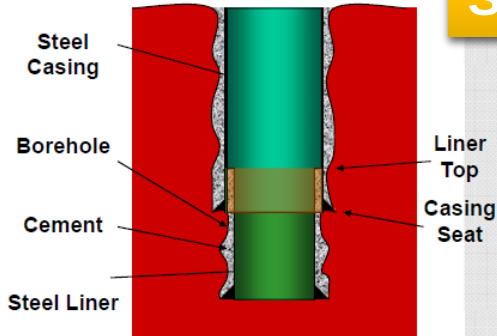
Remedy cementations



Cementation types

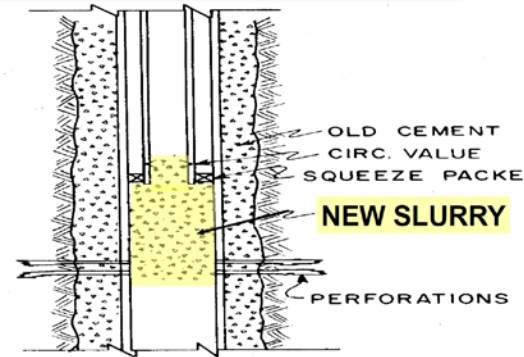


Full String Cementing

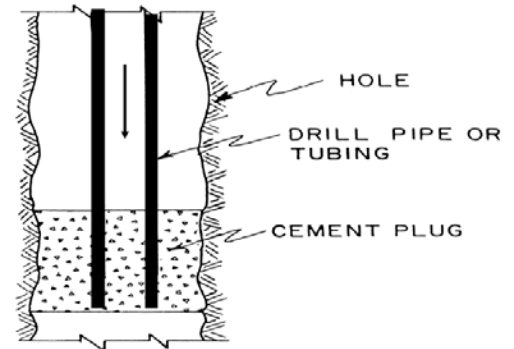


Liner Cementing

Squeeze cementing

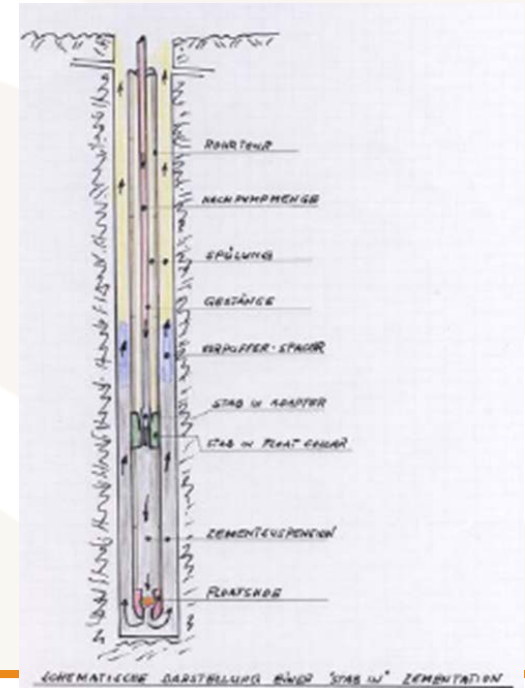
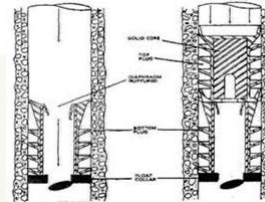
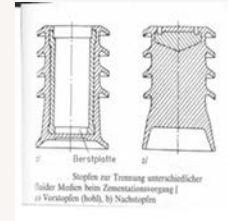
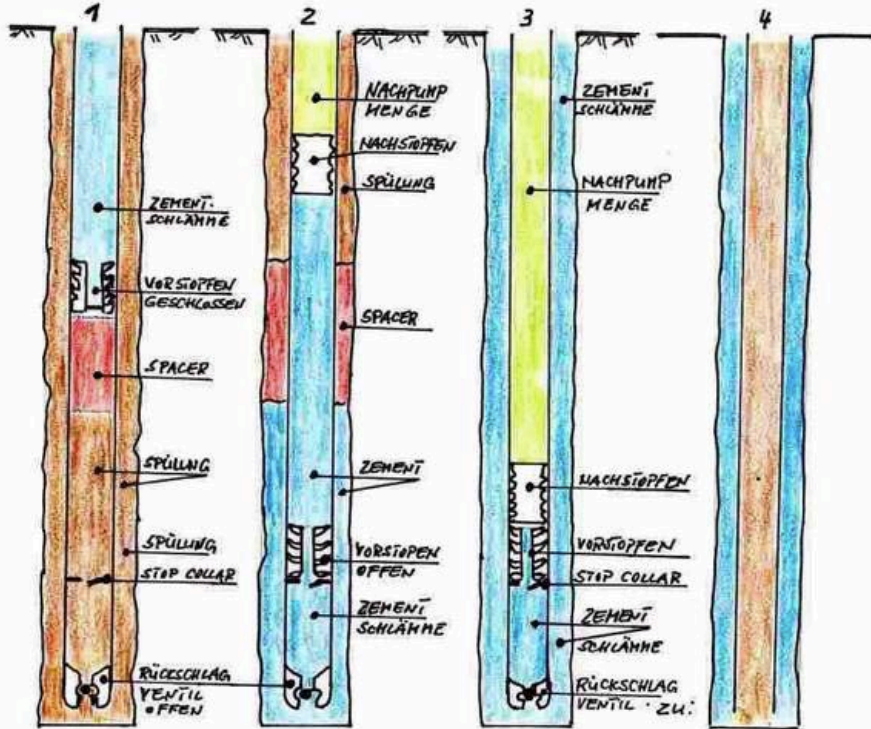


Cement plug



Bottom up cementing with plugs + stab-in

Stab-in cementing



Cementation planning

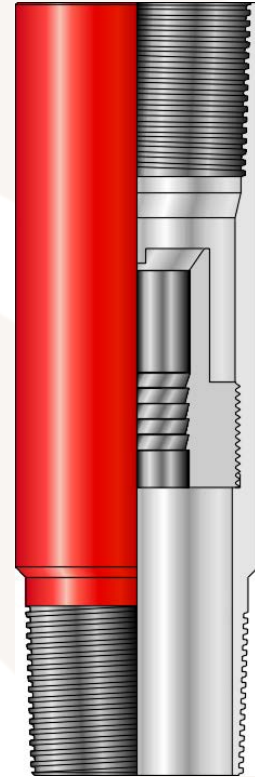
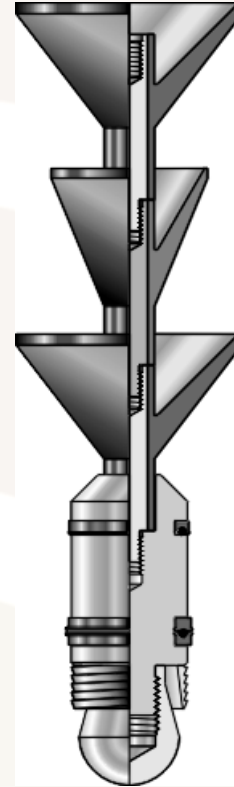
- Volume of cement slurry (borehole volume, casing volume etc.)
- Caliper measurement
- Density of cement slurry
- Recipe of cement slurry (cement, water, additives)
- Rheological properties and setting time
- Determine pump parameters (**pressure + pumping rate -> time !!**)

Casing cementation tools

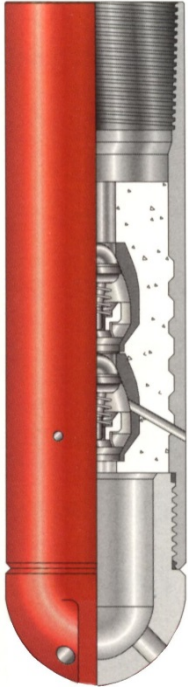
- Guide Shoe (casing shoe)
- Float Collar (check valve)
- Centralizer, Scratcher



Plugs



Float Shoe



Cementation equipment



Cement head

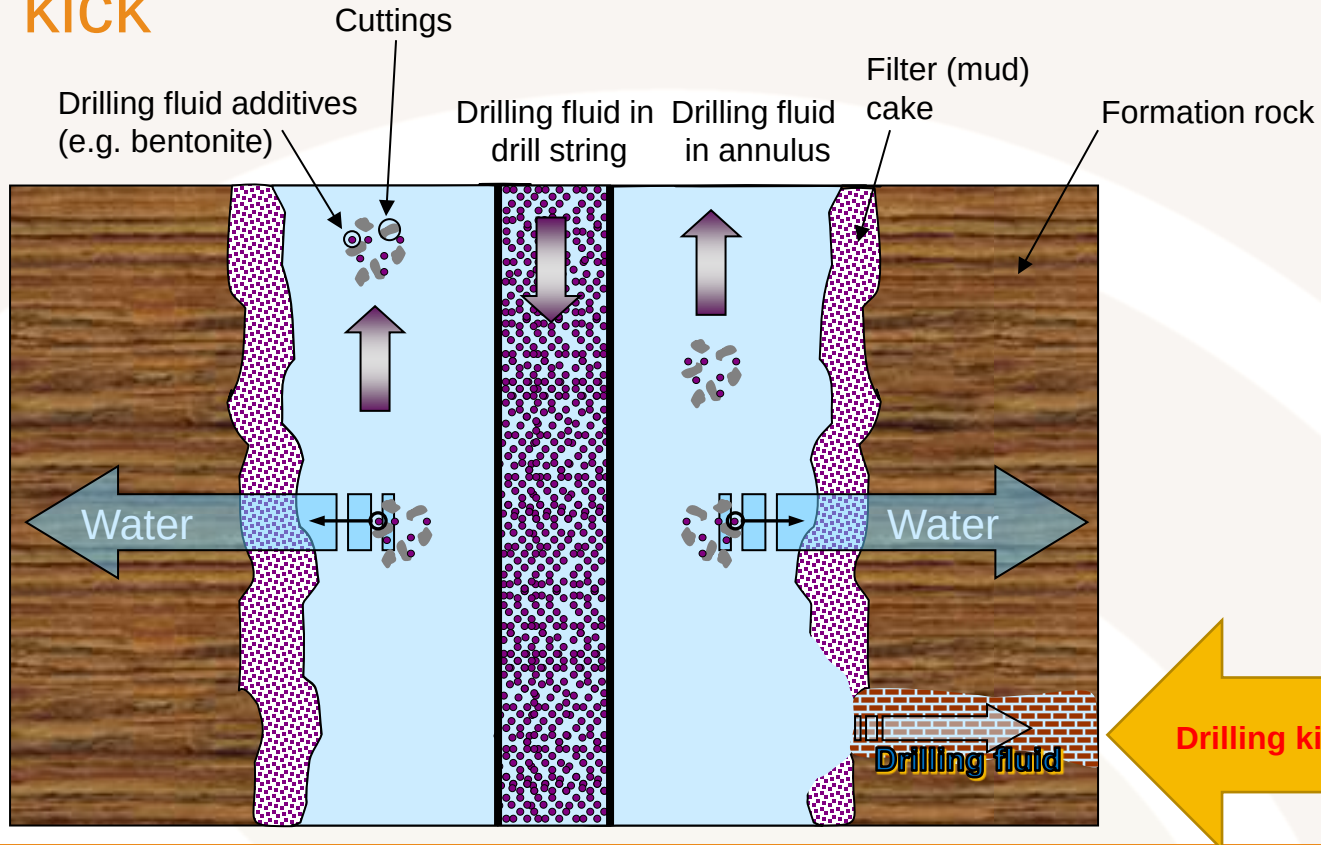


Cement delivery

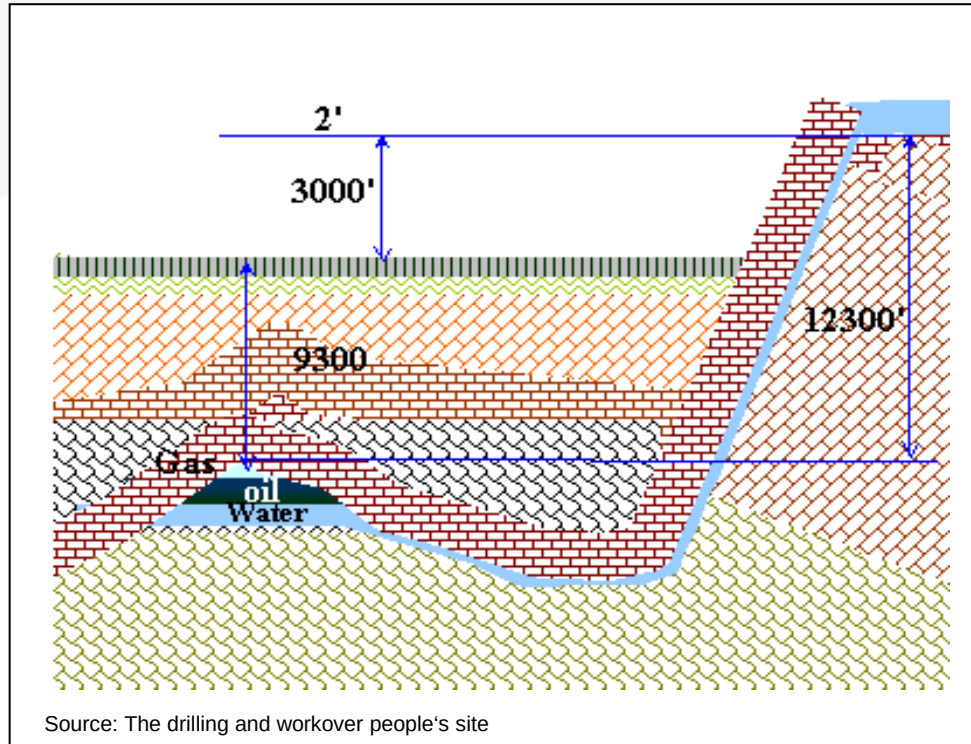


Well Control

Drilling kick



Fluid pressure - hydrostatic



**Hydrostatic overpressure
("Communicating pipes")**

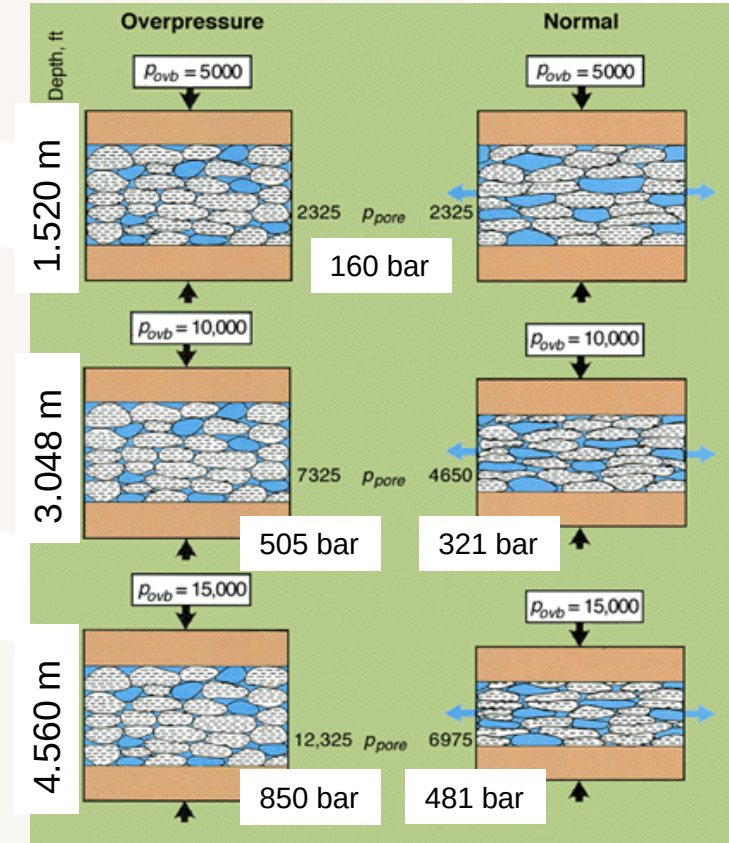
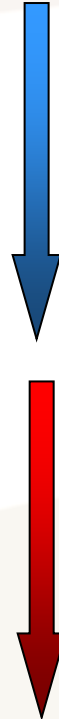
Fluid + pore pressure

Well drained conditions:

- Ballasted
- Increase of fluid pressure
- Decrease of pore space
- Fluid expulsion
- Compaction
- Normal hydrostat. pressure

Poorly drained conditions:

- ballasted
- Increase of fluid pressure
- "No" fluid expulsion
- "No" decrease of pore space
- "No" compaction
- excessive hydrostat. pressure



Source: Schlumberger Oilfield Glossary

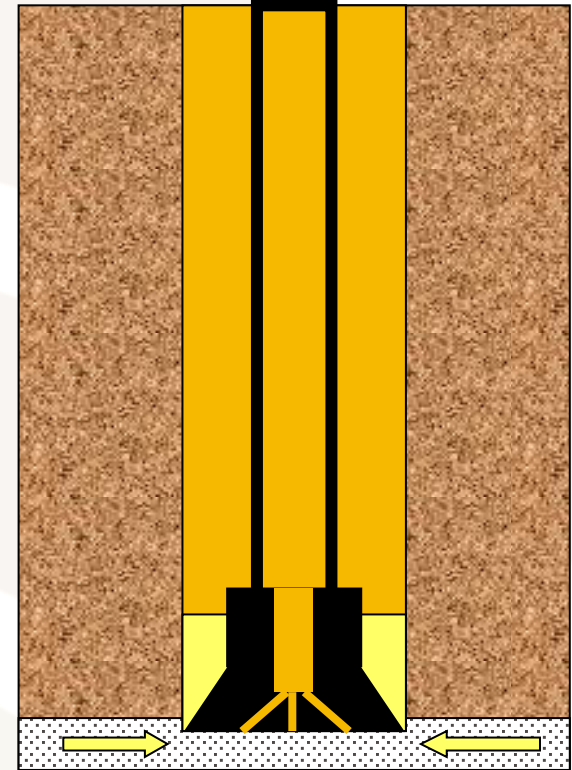
Well control

■ Kick while drilling

A flow of formation fluids into the wellbore during drilling operations. The kick is physically caused by pressure in the wellbore being less than formation fluid pressure, thus causing flow.

■ Blow-out

An uncontrolled flow of reservoir fluids into the wellbore, and sometimes catastrophically to the surface. A blowout may consist of salt water, oil, gas or a mixture of these.

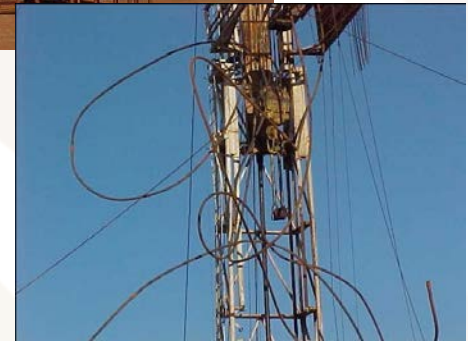


$$p_{\text{mud}} < p_{\text{fluid}}$$

Blow-out examples



Blow-out examples



Investigation of blow-outs over a period of 10 years

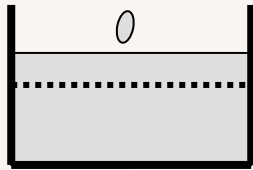
Reasons for the "initial kick":

- 42% insufficient drilling fluid level in the borehole
- 22% purge loss
- 16% swabbing
- 15% Insufficient drilling fluid density
- 5% other

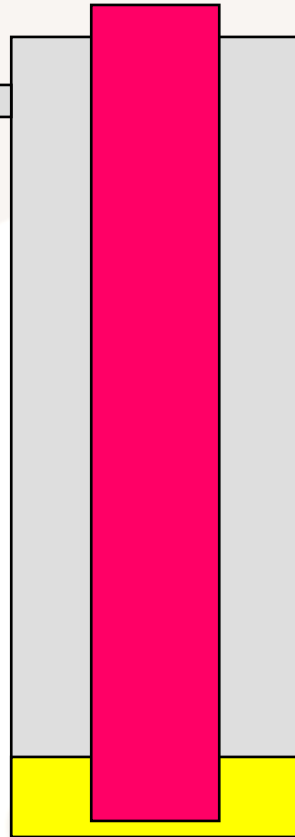
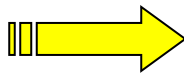


Kick Detection

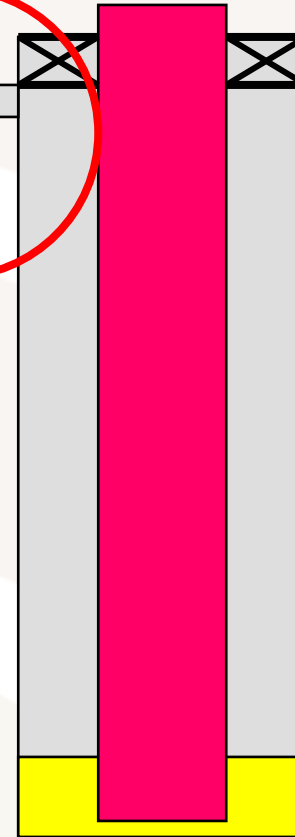
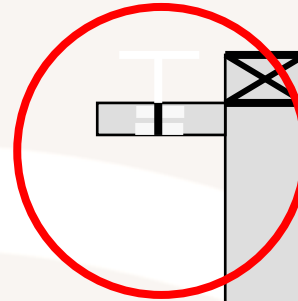
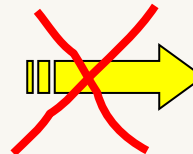
choke valve



$$P_{\text{formation}} > P_{\text{well}}$$



$$P_{\text{formation}} = P_{\text{well}}$$

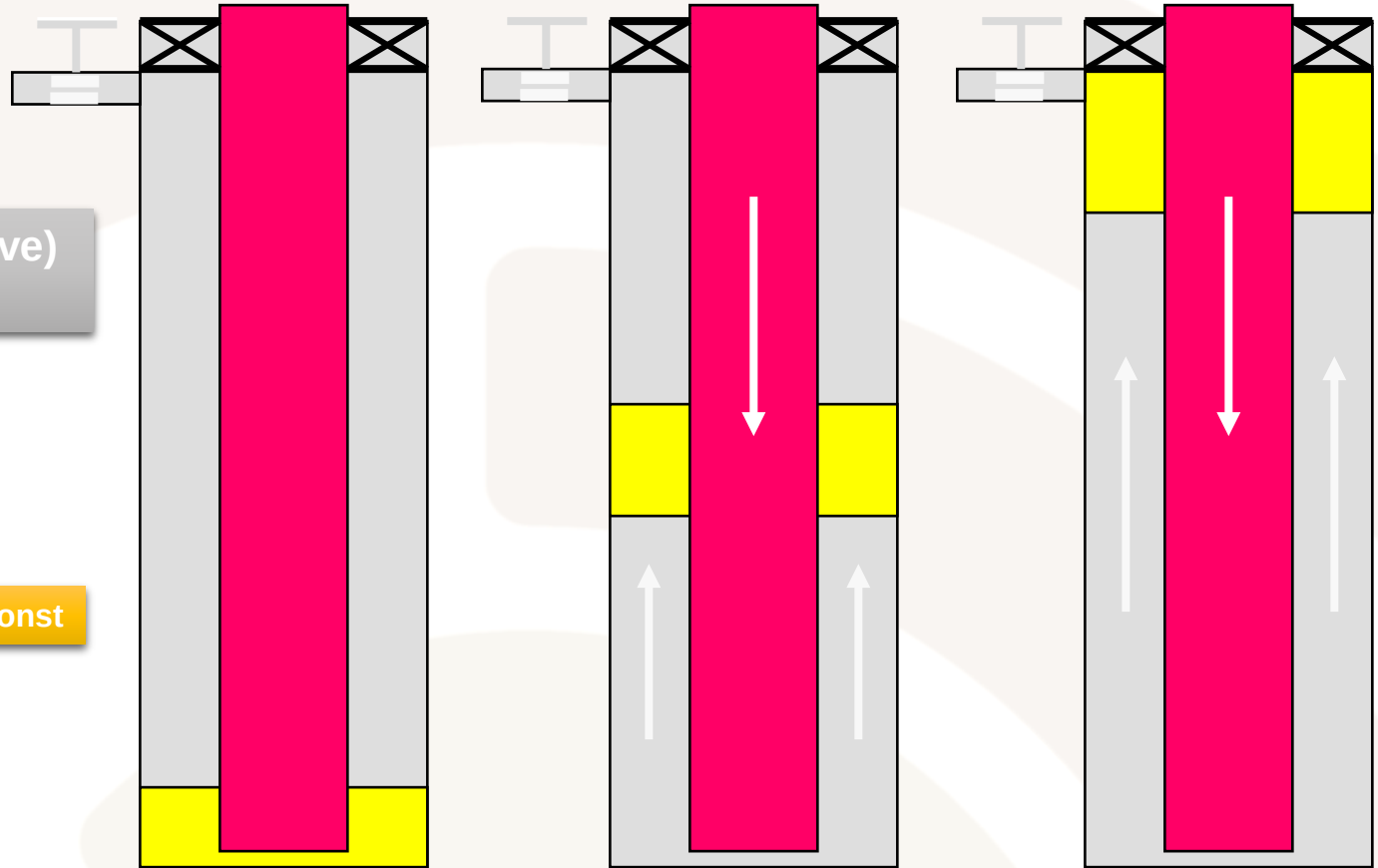


Kick Control

Close BOP
Switch off the pump

Circulate out (remove)
drilling kick

Bottom hole pressure = const



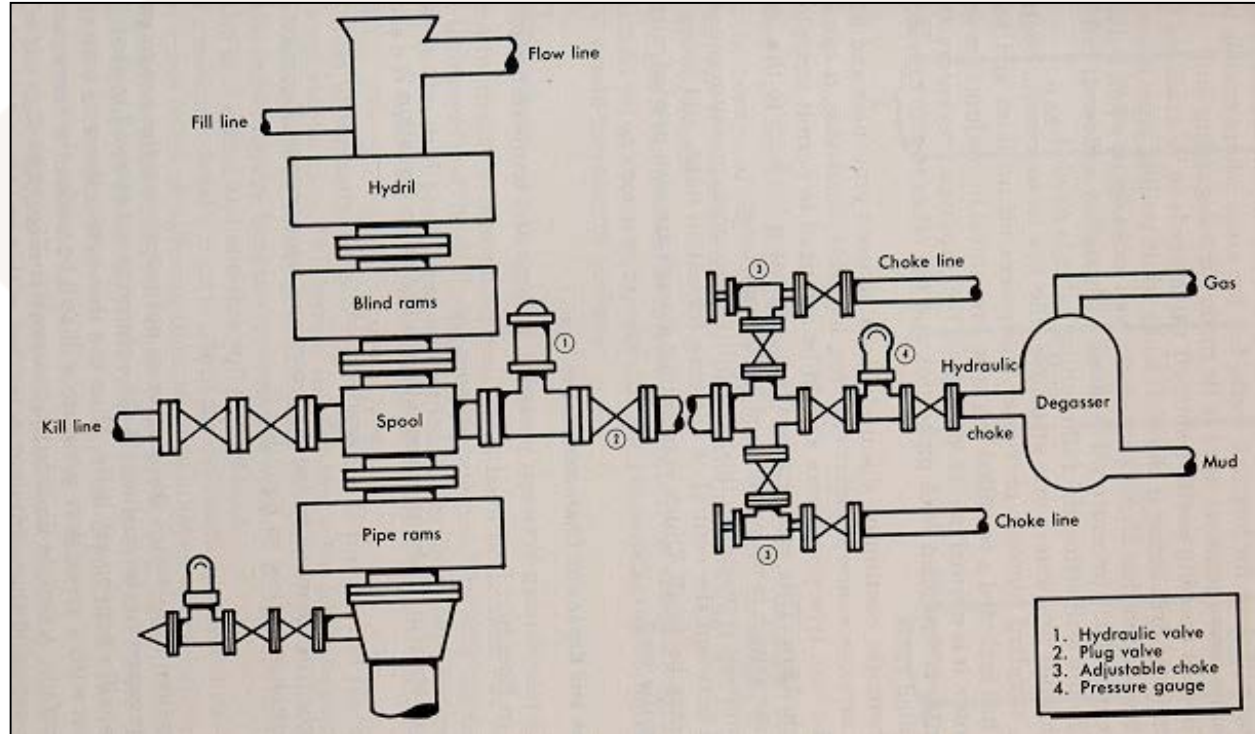
Methods of well killing

Killing a well: To regain control of a well after a kick or blow-out by recirculating high density mud

Methods of well killing:

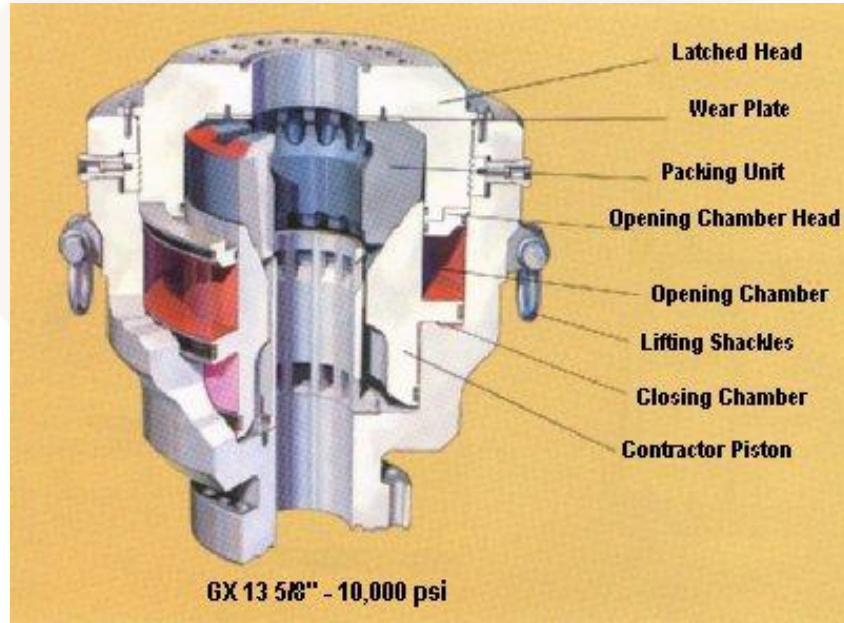
- Wait-and-Weight, Method. After the kick is shut in, weight the mud to kill density and then pump out the kick fluid in one circulation using the kill mud (also called *engineer's method*)
- Driller's method - successive weighting of the drilling fluid until drilling kick is over

Pressure Control Equipment – Blow Out Preventer



(Moore, P.L., 1974: p.305)

Annular preventer



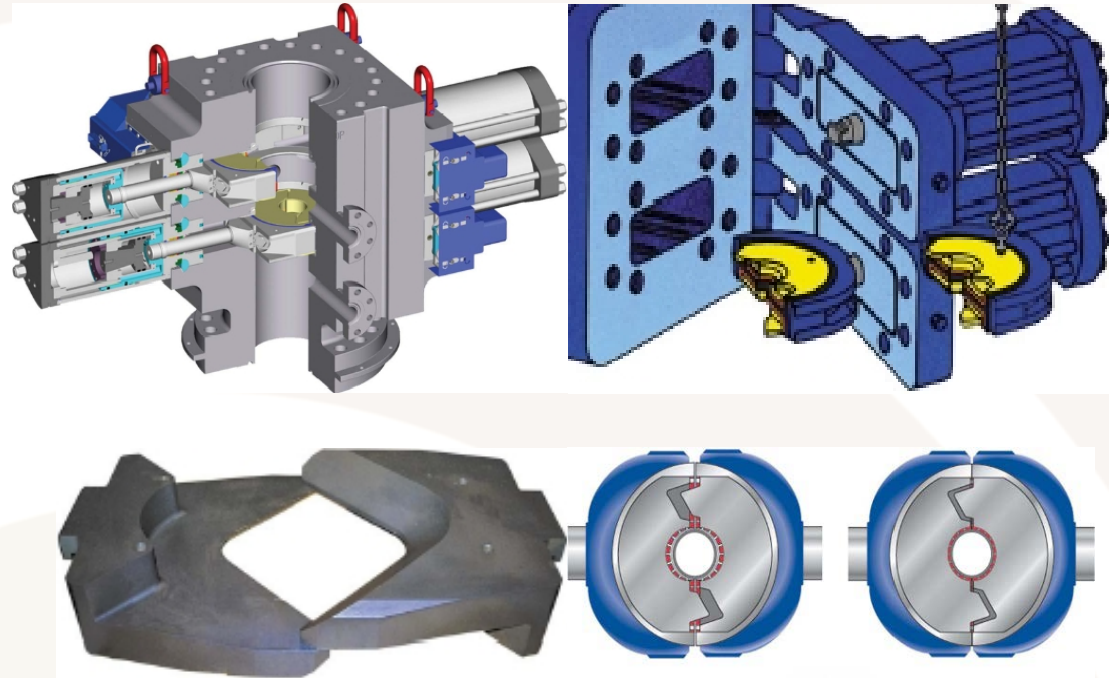
Closure of annular space or drill string space

Blow-out Preventer (BOP)

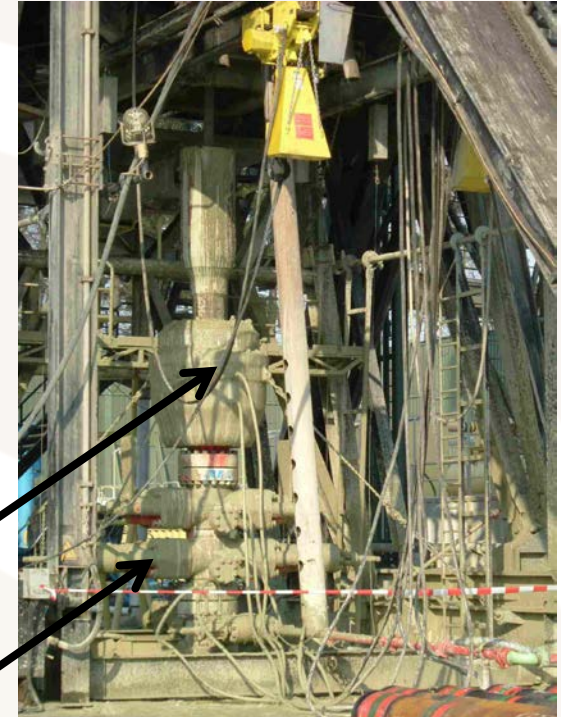
Pipe rams close around a drill pipe, restricting flow in the annulus (ring-shaped space between concentric objects) between the outside of the drill pipe and the wellbore, but do not obstruct flow within the drill pipe.

Blind rams have no openings for tubing, can close off the well when the well does not contain a drill string or other tubing, and seal it.

Shear rams are designed to shear the pipe in the well and seal the wellbore simultaneously. It has steel blades to shear the pipe and seals to seal the annulus after shearing the pipe.



Blow-out Preventer (BOP)



annular BOP

ram type BOP

Pressure Control Equipment

